

This will ensure independent scaling of individual services within an application with better resource utilisation.

Application life cycle automation tools like Ansible help in tracking, deploying and measuring the changes/enhancements in microservices.

Mozilla, Jboss, AngularJS and Bootstrap are the frequently used open source development tools used today. Other popular open source tools are Joomla, Jetty, OpenSSO, SOLR, Chef, Spring and WordPress.

Deployment architecture: The deployment platform chosen should be one that reduces the overall deployment complexity and supports open source based architectures. The suggestion is to consider using the dynamic component system for Java (e.g., OSGi). Component architectures enable a development model, whereby applications are dynamically composed of many different reusable components. These components hide their internals from other components and communicate through well-defined services.

Components can be installed, started, stopped, updated and uninstalled without bringing down the whole system. This can significantly reduce deployment times.

Integration: The value of a strong integration system lies in its ability to connect to a wide range of backend systems. It provides mediation, transformation, protocols and routing capabilities, apart from acting as a gateway to integrate with core back-end systems. It also provides aggregation and broker communications. Therefore, it is important to consider an integration solutions vendor that has the ability to connect with other databases, frameworks, applications and endpoints in a predictable manner.

It is important to go with open standards and open interfaces while choosing the integration backend as it should be free from vendor specific modifications and proprietary hooks. Mule, JBoss and WS02 are the famous integration products that can be used to establish the connected government.

Security: Effective security is the need of the hour. Security ensures that all the components of the cloud infrastructure are guarded against fast evolving threats. Applications and endpoints need to be protected and users granted access to applications as per the eligibility configured as part of the RBAC or role based access control.

Oauth2 and OpenID are a few of the open source security tools.

Customer experience: The customer experience is the product of an interaction between an organisation and a customer over the duration of their relationship. A good customer experience means that the individual's experience during all points of contact matches the individual's expectations. While building applications, it should be ensured that customer expectations are met through all interactions.

Analytics: A connected government may apply analytics to improve citizen services. Government decision management that is based on predictive analytics helps its services to be data driven.

Scikit-learn and Tensorflow are two popular open source machine learning tools.

SaaS: SaaS is a software licensing and delivery model in which software is licensed on a subscription basis and is centrally hosted. This model provides services without any requirement for customisation. It follows the twelve factors methodology to be portable and standardised. It is readily available and quick to take onboard.

Salesforce, AWS, etc, are popular open source SaaS.

PaaS: This is a category of cloud computing services that provides a platform allowing customers to develop, run and manage applications without the complexity of building and maintaining the infrastructure typically associated with developing and launching an application. This is the core platform on which cloud native applications and supporting systems are based.

Cloud Foundry, Cloudify and OpenShift are a few open source platforms.

By adopting OSS, governments become digital and more connected than ever before. They become FAST (flat, agile, streamlined and tech-savvy). OSS helps to drive public sector transformation and realise connected governance resulting in an enhanced quality of life.

OSS based next generation technologies enable the government to move from transaction-based services to relationship-based services. This also paves the way for new digital government services to be rapidly built, changed, enhanced and deployed, so that new ideas can be tried and refitted if needed. It also makes government data available openly, so that external parties can build services with adequate visibility of the government's functioning. **END** 

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